SESHADRI R

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PROFILE SUMMARY

- A competent professional offering several years **of experience** in **leading Software Engineering projects**, encompassing design, scope definition, process management and ensuring compliance with quality standards
- **Championed technology initiatives** to devise solutions harmonizing seamlessly with architectural principles, patterns, policies, and standards.
- **Steered architectural strategies** to promote the reuse of application architecture components, streamline technology, alleviate technical debt, and phase out legacy systems.
- **Ensured the strategic choice of relevant architectures,** principles, patterns, standards, and policies to adeptly address challenges and opportunities across business units.
- **Crafted comprehensive high-level product specifications,** fostering close collaboration with business units to discern both functional and non-functional prerequisites for cutting-edge software and applications.
- **Provided adept technical mentorship and coaching** to cross-functional teams of Developers and Engineers.
- **Harnessed machine learning and deep learning frameworks** to orchestrate the 5G ORAN Non and Near real-time controller, architecting ML workflows for orchestration, and overseeing end-to-end processes from training to deployment.
- **Innovated in designing Proof of Concept (PoC) initiatives**, leveraging machine learning and deep learning techniques alongside time-series data to bolster the capabilities of the 5G RAN Intelligent controller.
- Robust in **participating in Software Development Life Cycle (SDLC)** right from requirement analysis, documentation (functional specifications, technical design) to maintenance of the proposed applications
- Proven business acumen in managing **Service Delivery & Business Operations** functions and contributing higher rate of business growth

EDUCATION

- **Masters in Engineering in Electronics and Communication Engineering** from Bharathiar University, Coimbatore, specializing in Communication Systems
- **Bachelor of Engineering in Electronics and Communication Engineering** from Madurai Kamaraj University, Madurai

TRAININGS

- Undergone training in Linux Internals, C++, Core Java, Web Component Development (SCWCD)
- Certification in Hadoop Fundamentals, MapReduce Programming, Real-time data access using HBase and MongoDB (NoSQL DB) for Developers.
- Certificate in Blockchain Essentials from IBM Cognitive class.
- Certificate in Python 101 for Data Science from IBM Cognitive class.
- Agile Project Management from EdX, Agile Software Principles, Agile Development using Scrum Methodology.
- Introduction to Artificial Intelligence.
- Certification in Reactive Architecture - Principles, Domain Driven Design, and Microservices
- Certification in Cloud Native Application Design
- Certification in 5G ORAN Architecture Procedures and Use cases
- Undergoing courses on Deep learning using Tensor Flow 2.0, Pytorch and sci-kit learn
- **MLOps: Deploying AI&ML Models in Production using AWS**

CORE COMPETENCIES

IT Strategy Planning & Execution

Machine Learning & Deep Learning Frameworks

Solution Architecture & Design

Cloud Solutions & Deployment

Project Life Cycle Management

Agile and SCRUM Methodology

Process Excellence & Delivery

SDLC Processes

Team Management & Leadership

Multi Cloud Platform Architecture

Business Process Integration

PATENTS

- Filed 2 Software Patents in India and USA

Technology Transformation /Application Modernization/Solution Architecture

- Working knowledge on SysML, UML and MBSE tools like Magicdraw to create System Architecture and Requirements models.
- Applied Cloud Native design principles and patterns, leveraging Docker and Kubernetes Redhat Openshift platforms, and optimizing performance through Istio Service Mesh (Consul, Envoy)
- Implemented SOLID design patterns, cloud strategies and microservices patterns like CQRS and Saga.
- Devised scalable cloud platform solutions, prioritizing elasticity, performance, and resiliency while implementing service-based architectures
- Led customer-centric initiatives, generating impactful Proof of Concepts (POCs) and Minimum Viable Products (MVPs) on Docker, Kubernetes, and Redhat Openshift platforms
- Demonstrated proficiency in Event-Driven and Reactive Architecture, alongside Domain Driven Design (DDD) principles
- Shaped high-level solutions on AWS/Azure Cloud, with hands-on expertise in Core Java and exposure to J2EE
- Applied expertise in RedisDB and NoSQL databases such as MongoDB and Cassandra

Software /Technical Leadership

- Software Development Life Cycle (SDLC): Excellent knowledge of Software Development; skilled in gathering requirements & analysis, technical architecture, design & testing, mentoring & leading technical teams, quality & process improvements
- Application Development / Integration: Extensive global experience on custom application development, integration, systems analysis; resolved integration issues between various applications with focus on optimizing application performance
- Application Architecture: Experience in managing enterprise application architecture, designing, integrating and implementing Enterprise Applications, Product Development & POCs
- Agile Methodology: Facilitated the transition of projects from traditional methodologies to Agile – Scrum, Kanban and SAFe; steered Agile transformation efforts for digitalization and product development spend
- Global Exposure: Worked across multi-cultural IT environments, with global clients to deliver technically challenging projects
- Cross-functional Collaboration: Rich experience in dealing with Business Users, Technical Analysts, Domain Experts and Subject Matter Experts for knowledge transfers, transitions & delivery
- People Management: Directed productive cross-functional teams using motivational leadership that spurs people to give 100% effort

Domain Expertise : 5G Telecom/Wireless/Networking development on C++

- Led the development of C++-based 5G Cloud RAN software, including vRAN and O-RAN components, facilitating E2 interface and near-real-time gNB parameter control for ENDC, IMLB, and CA use cases
- Designed and implemented a distributed, high-performance C++11 and Java-based Near Real-time RAN Intelligent controller, serving the Telecom/5G RAN (gNB), ORAN-Near RT-RIC, Finance/Banking, and Healthcare domains
- Engineered a latency-sensitive application using C++11 /C++14 /C++17 and Java, catering to the specific requirements of Telecom/5G RAN (gNB) ORAN-Near RT-RIC, Finance/Banking, and Healthcare domains

AI/ML based O-RAN Use case /Solution Architecture design for 5g RAN

- Architecture to integrate AI/ML functionality within O-RAN systems
- AI/ML based RAN slice SLA Assurance use case detailed design
- AI/ML model and training pipeline design
- Analysis methodologies to evaluate the effectiveness of the AI/ML model AWS Sagemaker Model Monitor
- Analysis of Synthetic data generation options like GAN.

TECHNICAL SKILLS

- **Operating systems:** Sun Solaris and Windows NT/2000, Linux Red Hat 7.2, Real Time OS: VxWorks and OSE Delta
- **Tools & Utilities:** Microsoft Visual Studio tools and utilities, Advent Net C SNMP Agent development toolkit. Rational Rose **UML Tool. Tornado 5.1, CORBA (ACE TAO, Vertel, Orbix), Git, Gerrit, TensorFlow**
- **Programming Languages**: C, C++, Java , Python
- **Web Technologies:** Web component development, REST API design, WebServices
- **BigData:** Hadoop, MapReduce, HBase NoSQL, MongoDB, RabbitMQ, Storm, Flume
- **Network Protocols**: Understanding of Firewall(ACL,NAT,SSL) SS7, TCAP, INAP, SCTP
- **Cloud Platforms : AWS SageMaker, S3, ECR, EC2**

WORK EXPERIENCE

Oct 2023 – Present working at Witec as Senior Engineer
Project : Modular Control Architecture SW Platform Development

- Working on the various features of MCA platform and involved in analysis design and development of multithreaded C++ based

software
- Defining requirements through MagicDraw
- Good understanding of ML concepts and defining Use cases for automotive domain control system using ML.
- Familiar with Machine Learning algorithms,CNN,RNN for Deep Learning, Time Series forecasting, GAN & tools like Tensorflow.

Sr. Staff Software Engineer, Juniper Networks, Bangalore, Apr 2022 - 2023

Project :

AI/ML-based Non-RT RIC use case:

- Formulated a comprehensive high-level architecture/design for the 5G Network Slicing use case, with a focus on RAN Slicing
- Designed data ingestion pipelines and AI/ML training pipelines, including AI/ML model deployment and inference in the Non-RT RIC
- Created AI/ML-based RAN slice SLA use case, ensuring compliance with 3GPP standards and interworking with RAN NSSMF and NFMF
- Proposed architecture /design for real-time data ingestion pipeline from O1 interface (PM/FM) into AI/ML Workflow pipeline using Kafka and storage in TSDB.
- Designed AI/ML model training on DataBricks(DeltaLake) , storing models on AWS S3 bucket followed by deployment in Non-RT RIC as per O-RAN standards WG2
- Made use of AWS components/services like Sagemaker, Model Monitor, CloudWatch,AWS S3, EC2, and ECR for the PoC solution design.
- Enabled dynamic control of radio parameters using the E2 interface through the NearRT RIC.

5G ORAN RIC Platform feature development on Linux /C++.

- Developed features for ORAN-compliant RIC platform, focusing on various application microservices running on Kubernetes (K8s) and later on AWS
- Contributed to the enhancement of E2Scaling feature for RIC, specifically targeting improvements in E2M and E2T components.
- Designed and implemented the Subscription Manager statelessness feature
- Played a significant role in the design and development of subscription manager merging for RIC subscription
- Led the high-level design of the Statelessness feature across different microservices components within the RIC platform
- Conducted comprehensive analyses to bolster the RIC platform's stability, reliability, user experience, and application architecture

Service Management and Orchestration:

- Leading the analysis and design of a metrics collection framework for the Network Slicing use case on AWS within the near real-time RIC platform
- This involves mapping 5G Network slices' KPIs to provide an end-to-end view of the system health and usage patterns
- Utilized Prometheus and Grafana to design informative dashboards on AWS
- Explored the architecture of the Zero-touch Service Management (ZSM) for end-to-end Network Slicing and cross-domain service orchestration spanning RAN, Transport, and Core networks

May’20-Apr’22with NSN Bangalore as Software Architect

Last Project Executed: 5G RAN - gNB Control plane software Architecture and Design (SW Architect)

Key Result Areas:

- Engaged in the development of call control and Operations, Administration, and Maintenance (OA&M) C++ software architecture on the Redhat OpenShift Container Platform (OCP). Also orchestrated the solution design for the E2 Interface/agent, facilitating communication from the near real-time RIC (RAN Intelligent Controller) to both 5G-gNB and 4G LTE systems
- Led the formulation of requirement specifications for the E2 interface towards various components, including CU-CP, CU-UP, and DU
- Architected a unified C++ software architecture for E2, encompassing both standalone 5G-RAN and Cloud RAN scenarios, and effectively applied C++ Design patterns and SOLID Design principles
- Crafted the design for C++ based control plane and OA&M Software tailored to the specific requirements of gNB CU-CP, CU-UP, and DU, as well as application prerequisites
- Conducted an in-depth analysis and design of a C++ based E2 Agent container/Cloud strategy, implementing a microservices-based, scalable Cloud Native architecture to accommodate E2Agent and O&AM interfacing on AWS cloud
- Designed a scalable C++ based E2 interface within gNB, optimized to handle high throughput and latency demands associated with requests and responses

For more projects in the organization, refer annexure section

Feb’17-Oct’19 with CTS AS Principal Architect- Technology

Last Project Executed: 5G Cloud Radio Access Network Platform Development

- Worked on the design and development of multithreaded C++ based platform software infrastructure for 5G Cloud –Radio Acces Network
- Analysis of requirements High level architecture and design for 5G Cloud RAN on AWS

May'14-May'15 with EGSI Bangalore as Principal Software Engineer

Project : Generic Platform development ,SDN/NFV enablement using OpenFlow

OS : Linux (RHEL), VMware and KVM Language : C++/C SDN /NFV R&D

- Executed the separation of platform-independent and dependent code bases, enabling the operating system to seamlessly operate across multiple product
- Analyzed and designed the High Availability (HA) feature utilizing the openSAF-based Availability Management Framework for the common control plane
- Contributed to Quality initiative by implementing defect prevention measures, including enforcing static code analysis on IPOSref code

M2M/IoT using Cloud Systems and Data Analytics:

- Architected the software infrastructure for an analytics platform, specifically tailored to process real-time communication events within the Small Cell environment.
- Leveraged Apache Kafka and Storm as integral components of the platform's design to facilitate efficient data processing and analysis.

Aug'11-May'13: IBM India, Bangalore as Advisory Software Engineer

Project : Power Enterprise Systems/Server Virtualization Management | Environment: Linux ,Windows, KVM, Phyp using Java,C++ and C | Tools: GIT, Gerrit, CSNext RSA (Rational Software Architect)

Features Developed for Power Systems VM Management:

- Pioneered the design of a User Space File System using FUSE, tailored for Power Firmware Support Processo
- Enhanced the resource allocation optimization for Virtual Machines (VMs) within multicore sockets.
- Introduced LDAP-based user authentication support in the Power Management Console

RESTful WebServices API Design and Development for Power HMC:

- Led Software architecture and Object Model design to establish a REST API as the northbound interface for Power Systems Management
- Engineered the universally unique identification (UUID) and persistence framework (openJPA) for managed objects, ensuring reliable and coherent data management

Nov'06-Aug'11: Unisys India, Bangalore as Senior Technical Architect

Environment: Linux (SLES10 and RHEL5)

Language: C/C++, Virtualization technologies like Xen and VMWare ESX.

Tools Used: UML for modelling the software (Rational Software Architect)

Key Result Areas:

High Availability and Disaster Recovery for Enterprise Servers:

- Led the design and development of High Availability (HA) solutions using C/C++ and open-source tools, including Heartbeat 2.0 on SUSE Linux 10 and Red Hat Cluster Suite. Utilized virtualization technologies such as VMWare and Xen
- Engineered Load Balancing solutions using C/C++ on the Linux Virtual Server (LVS), optimizing scalability and load distribution

Enterprise Systems Management Tool Development:

- Served as the Technical Lead for the development of the RTI tool, taking end-to-end ownership of the product from conception to delivery
- Carved out the roadmap for a Data Center Management tool, specifically tailored for log collection and analysis within an enterprise environment

PREVIOUS EXPERIENCE

May'06-Nov'06: Multitech Software Systems India

Project: IP Enabling of Serial Devices/ NAT Router | **Environment:** Linux and C/C++ | **Role:** Project Manager/Engineering Lead | **Environment:** ARM,Linux, C

Dec'04-Mar'06: Cisco Systems, Bangalore and San Jose as Software Engineer IV (Sr.Software Engineer)

Project: L4-L7 Security Application Development | **Environment:** Linux and C/C++

Aug'02-Nov'04: ADC India Comm and Infotech Bangalore as Technical Leader

Project: SNMP Agent Development for LRCS and Software Defined Radio | **Environment :** Windows 2000 and Linux and C/C++

Mar'02-Aug'02: Larsen and Toubro Infotech Bangalore as Project Leader

Project: Mobile IPv6 Development and CDMA2000 BSS enhancements | **Environment:** Linux and C/C++

Apr'01-Feb'02: Ericsson Wireless Communications Inc, San Diego, CA as Senior Engineer

Project: CDMA2000 Base Station Controller Development,| **Environment:** OSE Delta and Sun Solaris, C++

Jan'99-Mar'01: Lucent Technologies, Bangalore as Lead Engineer

Project: UMTS RNC and Node B development for Bell Labs | **Environment:** Sun Solaris, VxWorks, C++

Aug'97-Jan'99: Robert Bosch India ,Bangalore as Senior Software Engineer
Project: SS7 based Intelligent Network,Nortel , UK| **Environment:** HP-UX, HP-SOS, Protel

May'96-Aug'97: Wipro Systems, Bangalore as Senior System Engineer
Project: MEDCOM Application Development | **Environment:** Windows NT , VC++ UML, Win32 API, COM/DCOM

Jul'95-May'96: Indira Gandhi Centre For Atomic Research, TN as Scientific Officer
Environment: Unix, c

ANNEXURE

Projects in NSN

5G – ORAN Near RT RIC Design and Development

- Designed and developed C++ software architecture for the Near RT RIC platform
- Identified and designed features, including Metrics and scalability, for Near RT RIC on the Kubernetes-based platform
- Developed C++ software architecture and roadmaps for features/use cases such as 5G RAN Slice SLA assurance, RAN Admission control xapp/rapp, 5G RAN NSSMF, and E2AP 2.0 enhancements
- Interfaced with O1 for Service Management and Orchestration

Feature Development for 5G LightSpan Fibre Devices

- Designed the high-level architecture of microservices-based Kubernetes cluster supporting a scalable and highly available 5G Access Controller platform and application services
- Acted as the Feature Owner for scalable Performance Monitoring (Bulk-IPFIX, Live (Netconf) Collectors) and Inventory Collector (Netconf, SNMP) components of the 5G Management Controller
- Created high-level architecture and design for Intent-Based Networking and Network Slicing for 5G devices
- Gained exposure to device yang models for Lightspan devices based on BBN Forum yang models
- Utilized software architecture design involving Apache Kafka, openTSDB, ElasticSearch, FluentD, Grafana, Zookeeper, and Prometheus
- Worked on features including circuit breaker for Inventory Collector, export of bulk PM data in a sliced environment, and slicing of Live Collector data for 5G Access Controller platform and 5G Network Access Virtualizer

Projects in CTS

Project: 5G Cloud Radio Access Network Platform Development

- Contributed to the design and development of multithreaded C++ based platform software infrastructure for 5G Cloud-Radio Access Network
- Analyzed requirements and created high-level architecture and design for 5G Cloud RAN

Project: Digital Engineering

- Analyzed high-level requirements of North American Healthcare providers and proposed re-architecting of existing software using Microservices, Event-driven, Reactive Architecture, and Domain-driven design (DDD) principles.
- Worked on a solution using Cloud Native design principles and proposed a solution/PoC using ServiceMesh for an event-driven, polyglot, scalable distributed system for migrating a legacy healthcare provider application.

Project: Modernization of Legacy Data Platform

- Devised a high-level architectural approach for batch and stream processing of data for the digital platfor
- Designed APIs and Management for the AWS cloud solution, considering tight SLAs
- Provided recommendations for data ingestion connectivity architecture
- Compared stream processing (Kafka/Kinesis) and processing engine (Spark/Samza)
- Proposed solutions considering Lambda/Kappa architecture for big data

Project:FRTB requirement analysis for a North American Bank

- Analyzed the FRTB (Fundamental Review of Trade Book) requirements and identified the Use Cases for Data Completeness, Data accuracy, Data Timeliness for implementing the Data Controls for the Risk IT .
- Proposed a solution for Standardized Approach (SA), specifically supporting SBA Calculator interfacing with the banks existing Data Layer based on Hadoop along with Metadata management.
- Provided a high-level cloud native architecture to host the solution of AWS cloud for scalability, elasticity and resilience.

Project: TEMENOS WealthSuite / TAP Refactoring:

- Analyzed high-level requirements of the TEMENOS TAP refactoring/modernization project and proposed a redesign of existing software for enhanced scalability, elasticity, resilience, fault-tolerance, and cloud deployment ease
- Designed microservices-based architecture and related APIs as a proposal/PoC for refactoring a monolithic application into a microservices-based architecture